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Learn How to Set Your \$PATH Variables Permanently in Linux

[Marin Todorov](#)

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In Linux (also UNIX) **\$PATH** is environment variable, used to tell the shell where to look for executable files. **\$PATH** variable provides great flexibility and security to the Linux systems and it is definitely safe to say that it is one of the most important environment variables.

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Programs/scripts that are located within the **\$PATH**'s directory, can be executed directly in your shell, without specifying the full path to them. In this tutorial you are going to learn how to set **\$PATH** variable globally and locally.

First, let's see your current \$PATH's value. Open a terminal and issue the following command:

```
$ echo $PATH
```

The result should be something like this:

```
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
```

The result shows a list of directories separated by colons. You can easily add more

directories by editing your user's shell profile file.



In different shells this can be:

1. Bash shell -> ~/.bash_profile, ~/.bashrc or profile
2. Korn Shell -> ~/.kshrc or .profile
3. Z shell -> ~/.zshrc or .zprofile

Please note that depending on how you are logging to the system in question, different file might be read. Here is what the bash manual says, keep in mind that the files are similar for other shells:

/bin/bash

The bash executable

/etc/profile

The systemwide initialization file, executed for login shell

~/.bash_profile

The personal initialization file, executed for login shells

~/.bashrc

The individual per-interactive-shell startup file

~/.bash_logout

The individual login shell cleanup file, executed when a

~/.inputrc

Individual readline initialization file

Considering the above, you can add more directories to the **\$PATH** variable by adding the following line to the corresponding file that you will be using:

```
$ export PATH=$PATH:/path/to/newdir
```

Of course in the above example, you should change **"/path/to/newdir"** with the exact path that you wish to set. Once you have modified your **.*rc** or **.*_profile** file you will need to call it again using the **"source"** command.

For example in bash you can do this:

```
$ source ~/.bashrc
```

Below, you can see an example of mine **\$PATH** environment on a local computer:

```
marin@[TecMint]:[/home/marin] $ echo $PATH

/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin
```

This is actually a good practice to create a local **"bin"** folder for users where they can place their executable files. Each user will have its separate folder to store his contents. This is also a good measure to keep your system secured.

If you have any questions or difficulties setting your **\$PATH** environment variable, please do not hesitate to submit your questions in the comment section below.

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I am a bachelor in computer science and a Linux Foundation Certified System Administrator. Currently working as a Senior Technical support in the hosting industry.

In my free time I like testing new software and inline skating.

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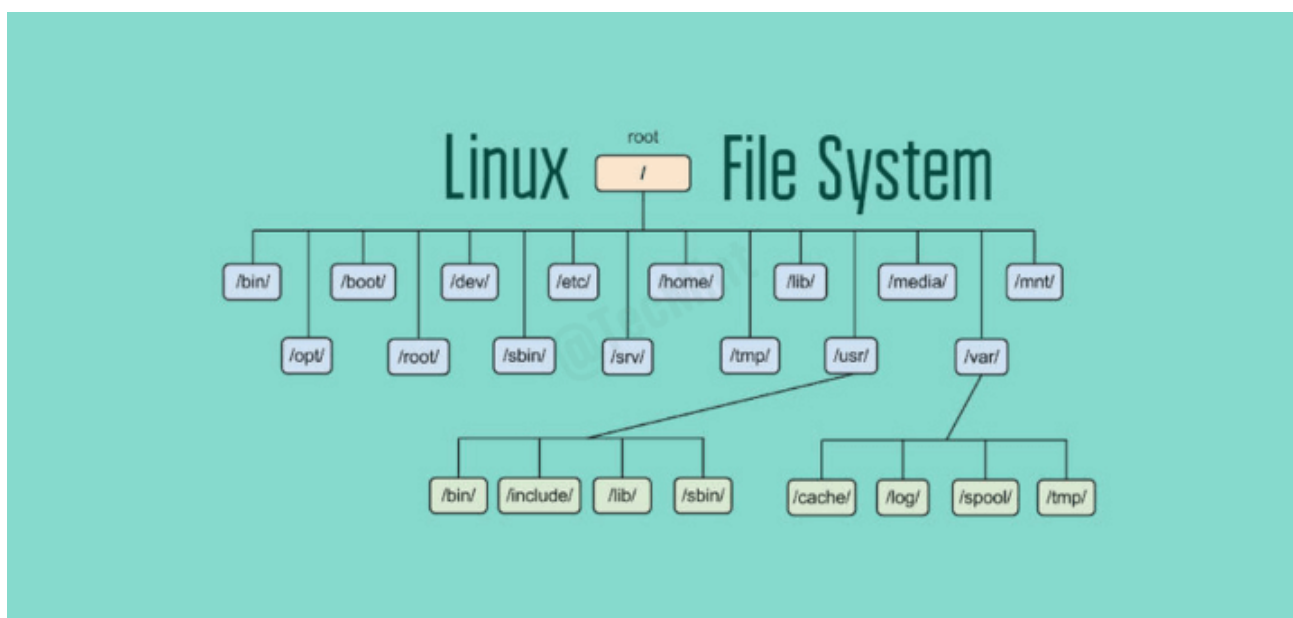
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```
[tecmint@TecMint:~]$ tar -xf tecmint.tar ravi2.txt
[tecmint@TecMint:~]$ ls -l ravi2.txt
-rw-rw-r-- tecmint tecmint 3.0 KB Thu Aug 17 13:37:30 2023 ravi2.txt
[tecmint@TecMint:~]$ vi ravi2.txt
[tecmint@TecMint:~]$ ls -l ravi2.txt
-rw-rw-r-- tecmint tecmint 7.0 KB Thu Aug 17 13:51:49 2023 ravi2.txt
[tecmint@TecMint:~]$ tar -uvf tecmint.tar ravi2.txt
ravi2.txt
[tecmint@TecMint:~]$ tar -tvf tecmint.tar
-rw-rw-r-- tecmint/tecmint 4015 2022-04-14 11:52 tecmint_monitor.sh
-rw-rw-r-- tecmint/tecmint 6789 2023-05-27 11:27 tecmint.json
-rw-rw-r-- tecmint/tecmint 297 2023-06-14 12:50 tecmint.py
-rw-rw-r-- tecmint/tecmint 0 2023-08-03 14:55 server.xconf
-rwxrwxr-x tecmint/tecmint 444 2023-06-14 12:57 tecmint.pl
```

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```
TecMint.com
tecmint@TecMint:~$
tecmint@TecMint:~$ yt-dlp -x --audio-format mp3 https://www.youtube.com/watch?v=wp430dtAAkM
[youtube] Extracting URL: https://www.youtube.com/watch?v=wp430dtAAkM
[youtube] wp430dtAAkM: Downloading webpage
[youtube] wp430dtAAkM: Downloading android player API JSON
[info] wp430dtAAkM: Downloading 1 format(s): 251
[dashsegments] Total fragments: 1
[download] Destination: Kate Bush - Running Up That Hill - Official Music Video [wp430dtAAkM].w
ebm
[download] 100% of 4.43MiB in 00:00:00 at 15.55MiB/s
[ExtractAudio] Destination: Kate Bush - Running Up That Hill - Official Music Video [wp430dtAAk
M].mp3
Deleting original file Kate Bush - Running Up That Hill - Official Music Video [wp430dtAAkM].we
bm (pass -k to keep)
tecmint@TecMint:~$
```

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16 thoughts on “Learn How to Set Your \$PATH Variables Permanently in Linux”

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Todd

[June 30, 2019 at 12:20 am](#)

@Ryan

When setting up the Android & Java SDKs on this system I had the same issue when defining the needed PATH variables and adding the library paths. I dug around and found that my current system has both `~/.bashrc` and `/etc/profile` and that no matter what I tried defining them in `~/.bashrc` & it wasn't working, but commenting the lines out in `~/.bashrc` and copying them to `/etc/profile` did exactly as expected.

I came to the conclusion that perhaps there's an overwrite happening, but I've never had a chance to understand it further.

[Reply](#)

Ryan

[June 29, 2019 at 6:21 am](#)

Hello,

I am trying to change my `$PATH` and i did a `vi` on `.bash_profile` where i added this command:

```
export PATH=$PATH:/HOME/scripts
```

Then I saved the file and then i did a `source ~/.bashrc`, but my scripts file did not show up when i ran this command.

```
$ echo $PATH
```

WHAT did i do wrong? Respectfully request your help.

Thank you.

cajunchief

[Reply](#)

Shima

[May 13, 2019 at 10:30 pm](#)

Hi,

My system is Fedora 28 and the default shell is **tcsh**. I want to know how can I set permanent PATH? It is necessary to say that I do not have root access.

Before, I had Ubuntu, the default shell was **BASH**. So I used to edit **.bashrc** file.

Thanks,

Shima

[Reply](#)

chk666

[June 4, 2018 at 3:58 am](#)

Hello and good days.

I need help what happened if i run **virtualenv** in my environment path, what i mean

is "**chk@waklu:\$ virtualenv**" did the path environment is change on **bashrc**, **profile**, or even in the **/etc/bash**, **/etc/profile**.

The problem is, I cannot detect my python and when i try to echo path should get **/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/home/marin/bin**

but I am getting blank, not showing up. Need help I run 16.04 Ubuntu.

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